

(MAN - codefest 08 :

Given $N=1024$, firing rate $f=50\text{Hz}$ (spikes per second).

Tasks

1. mean aggregate spike rate $R = N \times f$

$$\text{So } R = 1024 \times 50\text{Hz} = 51200 \text{ spikes/second}$$

2. $R=51200$ spikes/second, each AER Packet contains 10-bit neuron address = 1024.
each packet = 20 bits,

$$\text{So } B = R \times 20 = 1024000 \text{ bits/second}$$

$$\text{So } B \text{ in Mbits} = \frac{1024000 \text{ bit/second}}{1000000} = \underline{1.024 \text{ Mbits/second}}$$

3. So $B = 1.024 \text{ Mbits/second}$.

Compare it to each interface (I^2C , SPI, AXI4-lite).

So $I^2C \leq 3.4 \text{ Mbits/s}$, It can sustain $1.024^m \leq 3.4 \text{ Mbits}$

SPI $\leq 50 \text{ Mbits/s}$, It can sustain $1.024^m \leq 50 \text{ Mbits}$

AXI4-lite 100 Mbits/s , It can sustain $1.024 \frac{\text{Mbit}}{\text{sec}} \leq 100 \frac{\text{Mbit}}{\text{sec}}$

All 3 sustain the mean rate.

The lowest-complexity interface is I^2C as the one with the lowest max speed that still exceeds the B. because if it can handle the bandwidth & it's simplest protocol is simplest Protocol - 2 wires.

4. Burst Analysis.

25% of 1024 neurons fire within 1ms window = 256

Each packet = 20 bits

total bits in that burst = $256 \times 20 \text{ bits} = 5120 \text{ bits}$

in mbits = ~~0.00512~~ 0.00512 Mbits. per 0.001 sec $\Rightarrow 5120000 \text{ bits/sec}$
 $= 5.12 \text{ Mbits/sec}$

Burst to mean ratio is

Peak Mbits/sec $\div$ 1024 Mbits = $\frac{5.12}{1.024} = 5$
 the burst is 5x times heavier than mean,

~~Yes~~ J2C cannot handle the burst ($5.12 \text{ Mbits/sec} > 3.4 \text{ Mbits/sec}$)

The buffer needs to absorb the extra bits during the burst.

Burst Peak = 5.12 Mbits/s / J2C can drain = 3.4 Mbits/s .

Excess during 1ms burst = $\frac{(5.12 - 3.4) \text{ Mbits/s}}{0.001 \text{ s}} \times 10^{-3} = \frac{1.72 \times 10^3 \text{ Mbits}}{1}$

= 1720

= $1.720 \times 10^3 = \underline{1720 \text{ bits}}$ of buffering required.

5. A conventional system samples all 1024 neurons every 1ms, sending 1 bit per neuron per sample.

1ms interval = 1000 samples/second.

frame bandwidth = $1024 \text{ neurons} \times 1 \text{ bit} \times 10^3 \text{ samples/second}$

= 1.024 Mbit/sec .

AER to-frame ratio at $f = 50 \text{ Hz}$

$\frac{\text{AER bandwidth}}{\text{frame bandwidth}} = \frac{1.024}{1.024}$

$N \times f \times 20 = N \times 1000 \Rightarrow f = \frac{1000}{20} = \underline{50 \text{ Hz}}$

$f_{\text{crossover}} = 50 \text{ Hz}$, AER is the right choice when the mean firing rate is below 50 Hz.